
Computational Modeling of Ge/SiGe Heterostructures for Spin Qubit Arrays

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Abstract

Germanium heavy-hole spin qubits in strained Ge/SiGe quantum wells are emerging as one of the most promising routes to scalable quantum computing, yet every aspect of their performance—gate fidelity, qubit uniformity, and manufacturing yield—is locked in by heterostructure design decisions made long before the first device is ever measured. This report describes a complete, bottom-up computational design study covering six coupled physics layers: biaxial strain mechanics and Matthews–Blakeslee dislocation safety; heavy-hole–light-hole (HH–LH) splitting calculated from Bir–Pikus deformation potentials combined with finite-square-well quantum confinement; Monte Carlo manufacturing-tolerance and yield analysis; g -factor variability and qubit addressability; lab-frame quantum dynamics with quasi-static charge-noise dephasing; and a joint multi-objective optimisation that pulls everything together into a single figure of merit.

The recommended operating point is $x^* = 0.79$, $L^* = 8.3$ nm, which simultaneously delivers a 94 meV HH–LH splitting, a dislocation-safe margin of $L = 70\% h_c$, 99.96% single-qubit gate fidelity, and essentially 100% manufacturing yield under realistic epitaxial tolerances. The binding constraint turns out to be dislocation safety rather than any materials or coherence property — a finding that shifts engineering priority away from band-structure fine-tuning and toward the precision of the growth process itself.

Keywords: Ge/SiGe heterostructures, heavy-hole spin qubits, Matthews–Blakeslee critical thickness, Bir–Pikus deformation potential, yield engineering, quantum gate fidelity, multi-objective optimisation.

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1. Introduction

Quantum computing has the potential to transform how we tackle problems in chemistry, cryptography, and combinatorial optimisation, but turning that potential into reality requires qubit platforms that are simultaneously high-fidelity, long-coherent, and manufacturable at scale. Semiconductor spin qubits check that last box more convincingly than most alternatives, because they sit naturally within the same epitaxial growth and lithographic patterning infrastructure already deployed for classical CMOS [1]. Among the many semiconductor spin-qubit variants, germanium heavy-hole (HH) qubits in strained Ge/Si_{1-x}Ge_x quantum wells (QWs) have attracted growing attention for three practical reasons.

First, spin-orbit coupling in the Ge valence band is strong enough to enable fully electrical qubit control through electric-dipole spin resonance, which removes the need for on-chip micromagnets that become increasingly difficult to route as qubit density grows [2]. Second, the *p*-like symmetry of the heavy-hole wavefunction causes it to vanish at nuclear sites, which strongly suppresses the hyperfine interaction and pushes dephasing times T_2^* into the microsecond regime even in natural-abundance germanium [3, 4]. Third, several industrial groups — IMEC, Intel, and Hitachi among them — have already demonstrated Ge/SiGe qubit fabrication on 300 mm wafers, making the path to large-scale integration at least plausible.

What makes this platform genuinely challenging is that the qubit’s observable performance is largely fixed at the design stage, before a single device is measured. Two structural parameters drive almost everything: the Ge fraction x of the Si_{1-x}Ge_x virtual substrate, which sets the biaxial compressive strain in the Ge well and therefore the primary HH-LH splitting; and the quantum-well thickness L , which adds a confinement-driven splitting contribution while simultaneously governing both the dislocation-safety margin and the charge-noise sensitivity of the qubit. Every downstream observable — *g*-factor, Larmor frequency, gate fidelity, manufacturing yield — flows from that one joint choice of (x, L) .

2. Literature Review

2.1. The rise of Ge/SiGe spin qubits

The Ge/SiGe qubit field moved fast in the early 2020s. The landmark result came from Hendrickx et al. [5], who demonstrated a four-qubit germanium quantum processor with two-qubit gate fidelities above 99%, conclusively showing that Ge hole spins can be coherently coupled through exchange interactions at fidelities approaching fault-tolerance thresholds. Around the same time, Lawrie et al. [3] systematically characterised spin relaxation times and individual qubit addressability in Ge quantum dots, providing the experimental grounding for the *g*-factor and addressability analysis carried out in this work. Scappucci et al. [1] later set out a comprehensive review of the germanium quantum information route, which remains the clearest single account of where the field stands and what the main open challenges are. A particularly relevant enabling result came from Lodari et al. [4], who showed that lightly strained Ge quantum wells can reach hole mobilities exceeding $10^6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, confirming that disorder in well-optimised Ge/SiGe heterostructures is low enough to support microsecond coherence times.

2.2. Strain mechanics and Matthews–Blakeslee theory

When a Ge layer is grown pseudomorphically on a Si_{1-x}Ge_x virtual substrate, it is forced to adopt the substrate’s smaller in-plane lattice constant, building up biaxial compressive strain. This can continue only up to the Matthews–Blakeslee (MB) critical

thickness h_c [6]: above h_c , the elastic energy stored in the layer exceeds the cost of forming a misfit dislocation, and the film partially relaxes through dislocation glide on $\{111\}$ slip planes. Those dislocations degrade both carrier mobility and qubit coherence, so staying well below h_c is a hard requirement in any practical design. The MB equation is implicit in h_c and depends on the mismatch strain f , Poisson ratio ν , and Burgers vector b ; for the dominant 60° mixed-character dislocations in Ge/SiGe, all of these are reliably tabulated [7]. Lattice constants follow Vegard’s law, whose linear interpolation between the Si and Ge end points has been well validated across the full Si–Ge composition range.

2.3. Valence-band physics: Bir–Pikus and Luttinger models

The strain-induced HH–LH splitting is described by Bir–Pikus (BP) deformation potential theory [8], which treats the coupling between the $J = 3/2$ valence manifold and the strain tensor perturbatively. Under biaxial compression in the (001) plane, the BP interaction pushes the HH state below the LH state by $\Delta E_{\text{strain}} = 2|b_v|\varepsilon_{\text{bi}}$, where b_v is the shear deformation potential and $\varepsilon_{\text{bi}} = \varepsilon_\perp - \varepsilon_\parallel$ is the deviatoric (symmetry-breaking) component of the strain tensor. This splitting is what isolates the HH as a well-defined two-level qubit system.

Quantum confinement in a finite well adds a second positive contribution through the difference in HH and LH zero-point energies. The relevant effective masses are set by the Luttinger Hamiltonian [2]; their values for Ge are taken from the compilation of Vurgaftman et al. [7]. It is worth stressing a subtlety that often gets glossed over in the literature: the infinite-well analytic formula significantly overestimates the LH confinement energy at experimentally relevant well thicknesses, because the unusually small LH out-of-plane mass ($m_{\text{LH},z} \approx 0.046 m_e$) allows substantial wavefunction leakage into the SiGe barrier. A proper finite-well calculation is therefore not optional—it is the only way to get quantitatively reliable results.

2.4. g -Factor physics and qubit addressability

The parallel g -factor of a heavy-hole spin in a strained Ge well has been characterised both theoretically [2] and experimentally [3], with reported values spanning $g_\parallel \approx 6$ –15 depending on strain and well thickness. This large, tunable g -factor is one of the platform’s practical advantages, enabling GHz Larmor frequencies at modest magnetic fields and therefore fast gate operations. The trade-off is qubit-to-qubit variability: if $g_\parallel$ varies too much across an array, microwave pulses can no longer address individual qubits without crosstalk. Veldhorst et al. [9] established the general addressability criterion that the Larmor frequency spread must substantially exceed the Rabi bandwidth, which translates into a concrete bound on σ_g/g at a given drive strength.

2.5. Gate fidelity and charge noise

In Ge hole-spin qubits, the dominant decoherence mechanism is $1/f$ charge noise coupled through the spin–orbit interaction, which shifts the qubit frequency on a timescale much longer than a single gate [3]. This quasi-static character produces a Gaussian free-induction decay $\sim \exp(-(t/T_2^*)^2)$ rather than the exponential that would result from Markovian (white) noise. The distinction matters enormously for fidelity estimates: a Lindblad master equation with σ_z dephasing, which is the standard tool for Markovian noise, overestimates the infidelity by orders of magnitude. Reaching the surface code threshold of roughly 99% [9] therefore requires choosing the right noise model as much as it requires good materials.

2.6. Multi-objective design and sensitivity analysis

Global sensitivity analysis [10] offers a systematic way to decompose output variance into contributions from each input uncertainty, making it clear which tolerances are worth tightening and which are essentially irrelevant. When several objectives compete as they do here, where improving the HH–LH splitting tends to push the design toward higher strain and thinner wells, both of which carry other penalties — Pareto-front analysis provides a natural language for describing the trade-off without forcing an arbitrary weighting at the outset. These tools are standard in engineering design but are not yet routine in the qubit-device literature; applying them systematically is one of the contributions of this study.

2.7. Motivation and Objectives

A closer look at the existing literature reveals that while significant progress has been made in Ge/SiGe spin-qubit experiments, the design methodology still lacks a unified framework. Strain engineering, band-structure calculations, yield estimation, g -factor modeling, and gate fidelity have all been treated as separate problems rather than interconnected pieces of a single design puzzle. This fragmented approach may be acceptable for a single qubit, but becomes increasingly problematic as the field moves toward large-scale arrays, where small per-qubit imperfections accumulate and collectively degrade system-level performance. The present study is motivated by this gap and aims to bridge it by applying Monte Carlo tolerance propagation, variance decomposition, and Pareto-front analysis, tools well established in engineering and economics to the Ge/SiGe qubit design context. Specifically, the objectives are pursued in sequence: mapping the strain tensor and dislocation-safety boundary; computing the HH–LH splitting with finite-barrier corrections; generating a two-dimensional yield map from realistic tolerances ($\sigma_x = 0.005$, $\sigma_L = 0.5$ nm); modeling g -factor spread and addressability; evaluating gate fidelity under physically accurate charge-noise conditions; and finally consolidating all results into a joint figure of merit to locate and validate the optimal design point.

3. Computational Methodology

3.1. Strain tensor and Matthews–Blakeslee critical thickness

The substrate lattice constant is obtained from Vegard’s law,

$$a_{\text{Si}_{1-x}\text{Ge}_x}(x) = (1 - x) a_{\text{Si}} + x a_{\text{Ge}}, \quad (1)$$

with $a_{\text{Si}} = 5.431$ Å and $a_{\text{Ge}} = 5.658$ Å [7]. The biaxial strain components in the pseudo-morphically grown Ge layer follow from the free-surface boundary condition ($\sigma_{zz} = 0$):

$$\varepsilon_{\parallel} = \frac{a_{\text{Si}_{1-x}\text{Ge}_x} - a_{\text{Ge}}}{a_{\text{Ge}}}, \quad \varepsilon_{\perp} = -\frac{2C_{12}}{C_{11}} \varepsilon_{\parallel}, \quad \varepsilon_{\text{bi}} = \varepsilon_{\perp} - \varepsilon_{\parallel}. \quad (2)$$

The Matthews–Blakeslee critical thickness is found by solving the implicit equation [6]

$$h_c = \frac{b(1 - \nu \cos^2 \theta)}{8\pi f(1 + \nu) \cos \lambda} \left[\ln \left(\frac{h_c}{b} \right) + 1 \right] \quad (3)$$

numerically at each value of x using Brent’s method. Here $f = |\varepsilon_{\parallel}|$, $b = a_{\text{Ge}}/\sqrt{2}$ is the Burgers vector magnitude, and $\theta = \lambda = 60^\circ$ for the dominant mixed-character dislocations on $\{111\}$ planes. A safety margin of $L < 0.7 h_c$ is imposed as a hard constraint throughout the design space.

3.2. HH–LH splitting: Bir–Pikus and finite-well confinement

The total splitting is written as the sum of two independently computed contributions,

$$\Delta E_{\text{total}}(x, L) = \underbrace{2|b_v|\varepsilon_{\text{bi}}}_{\text{Bir–Pikus strain}} + \underbrace{\Delta E_{\text{QC}}(L)}_{\text{finite-well confinement}}, \quad (4)$$

with deformation potential $b_v = -2.16 \text{ eV}$ for Ge [8]. The confinement term ΔE_{QC} is obtained by numerically solving the one-dimensional Schrödinger equation in a finite square well of depth $V_0 \approx 150 \text{ meV}$ using a finite-difference discretisation on $N = 2000$ grid points spanning $\pm 5L$, with eigenvalues extracted by Lanczos iteration. Separate calculations for the HH ($m_{\text{HH},z} = 0.20 m_e$) and LH ($m_{\text{LH},z} = 0.046 m_e$) give $\Delta E_{\text{QC}} = E_0^{\text{LH}} - E_0^{\text{HH}}$ [2]. The analytic infinite-well formula

$$\Delta E_{\text{QC},\infty} = \frac{\hbar^2 \pi^2}{2L^2} \left(\frac{1}{m_{\text{LH},z}} - \frac{1}{m_{\text{HH},z}} \right) \quad (5)$$

is computed alongside for comparison, and its deviation from the finite-well result directly measures the wavefunction-leakage correction.

3.3. Manufacturing tolerance and yield analysis

Yield is defined as $Y(x, L) = P(\Delta E_{\text{total}} > 50 \text{ meV})$, where the 50 meV threshold ensures a sufficiently pure HH ground state for qubit operation. Manufacturing variability is characterised by $\sigma_x = 0.005$ (XRD-calibrated CVD) and $\sigma_L = 0.5 \text{ nm}$ (MBE with in-situ rate calibration). Two complementary methods are used to evaluate yield.

Vectorised Monte Carlo: $N = 50\,000$ samples are drawn from $\mathcal{N}(\mu_x, \sigma_x^2) \times \mathcal{N}(\mu_L, \sigma_L^2)$ and propagated through the full physics chain of Eq. (4). The finite-well confinement energy $\Delta E_{\text{QC}}(L)$ is pre-tabulated on a 200-point grid and linearly interpolated inside the Monte Carlo loop, giving a roughly $10^4 \times$ speed-up with negligible loss of accuracy.

Gaussian linearisation: a first-order Taylor expansion around the nominal point yields $\sigma_{\Delta E}^2 \approx (\partial \Delta E / \partial x)^2 \sigma_x^2 + (\partial \Delta E / \partial L)^2 \sigma_L^2$, from which yield follows analytically as $Y = \Phi((\mu_{\Delta E} - 50 \text{ meV}) / \sigma_{\Delta E})$. This enables evaluation of the full 2-D yield map in milliseconds and is validated against the Monte Carlo to better than 1% everywhere.

Variance is decomposed into per-parameter contributions $(\partial \Delta E / \partial p \cdot \sigma_p)^2$ (the tornado chart) to identify the dominant manufacturing tolerance [10]. The epistemic uncertainty on the deformation potential ($\sigma_{b_v} = 0.10 \text{ eV}$, reflecting scatter in reported literature values) is deliberately excluded from the yield calculation, it describes uncertainty about a fixed material constant, not device-to-device variation, but is reported separately for completeness.

3.4. g -Factor model and addressability

The parallel and perpendicular g -factors are modelled at leading order in the HH–LH mixing ratio $\beta / \Delta E$ [2]:

$$g_{\parallel} \approx 6\kappa \left(1 - \frac{\beta}{\Delta E} \right), \quad g_{\perp} \approx \frac{\alpha_{\perp}}{\Delta E}, \quad (6)$$

with $\kappa = 3.41$ (the Ge Luttinger–Zeeman parameter), $\beta = 30 \text{ meV}$, and $\alpha_{\perp} = 12 \text{ meV}$. These two parameters are calibrated to reproduce the experimentally observed ranges $g_{\parallel} \approx 6\text{--}15$ and $g_{\perp} \approx 0.05\text{--}0.30$ [3]. The Larmor frequency is then $\omega_L = g_{\parallel} \mu_B B / \hbar$.

Variability in $g_{\parallel}$ comes from three sources propagated independently through Eq. (6): substrate composition noise σ_x , well thickness noise σ_L , and chip-to-chip fluctuations in the

Luttinger renormalisation ($\sigma_\kappa/\kappa = 1.5\%$). Addressability requires that the typical Larmor detuning between neighbouring qubits, σ_ω , comfortably exceed the π -pulse bandwidth Ω_R/π [9].

3.5. Quantum dynamics and single-qubit gate fidelity

With the qubit modelled as a two-level system, the lab-frame Hamiltonian is

$$H(t) = \frac{\hbar\omega_L}{2} \sigma_z + \hbar\Omega_R \cos(\omega_L t) \sigma_x. \quad (7)$$

Three levels of simulation are run in parallel. Full lab-frame integration of Eq. (7) uses adaptive RK45 with tolerances $\text{rtol} = 10^{-9}$, $\text{atol} = 10^{-11}$, explicitly resolving the ~ 84 GHz carrier and confirming that the rotating-wave approximation (RWA) envelope emerges naturally. A rotating-frame simulation on the time-independent Hamiltonian $H_{\text{rot}} = (\hbar\Delta/2)\sigma_z + (\hbar\Omega_R/2)\sigma_x$ is run with the same integrator for speed. To model charge noise correctly, $1/f$ dephasing is treated quasi-statically [3]: a random detuning $\delta\omega_L$ is drawn per shot from $\mathcal{N}(0, 2/(T_2^*)^2)$ and the density matrix is averaged over 2000 shots. This reproduces the physically correct Gaussian free-induction envelope and the analytic fidelity $F = 1 - (t_\pi/T_2^*)^2$. The Markovian Lindblad model with σ_z dephasing is deliberately not used, as it corresponds to white-noise dephasing and would overestimate infidelity by orders of magnitude for the $1/f$ noise characteristic of Ge hole qubits.

3.6. Joint multi-objective optimisation

The capstone analysis combines all preceding results into a single scalar figure of merit,

$$\text{FoM}(x, L) = Y(x, L) \times F_{\text{single}}(x, L)^N \times \exp\left(-\frac{\sigma_g/g}{\sigma_{\text{target}}}\right), \quad (8)$$

where $N = 100$ qubits and $\sigma_{\text{target}} = 3\%$. Two hard constraints kill the FoM entirely in infeasible regions: $L > 0.7h_c$ (dislocation risk) and $\mu_{\Delta E} < 50$ meV (insufficient HH purity). The optimum is found by exhaustive grid search over 220×140 points. Independently of the FoM, a Pareto front is computed in the $(Y, 1 - F)$ plane by ranking all feasible points without weighting, which exposes the yield–fidelity trade-off free of any parametrisation artefacts. Robustness of the optimum is checked by repeating the search under four tolerance perturbation scenarios: $\frac{1}{2} \times$ tighter, $2 \times$ looser, σ_x -only tightened, and σ_L -only tightened.

4. Results and Discussion

4.1. Strain tensor and dislocation safety

Figure 1 summarises the strain-tensor results. As expected from Vegard’s law, the substrate lattice constant grows linearly from a_{Si} to a_{Ge} with increasing x . The mismatch between substrate and Ge layer grows as x decreases (more Si, larger mismatch), driving $\varepsilon_{\parallel}$ more compressive and ε_{bi} larger. Both trends are favourable for HH–LH splitting, but they also reduce h_c steeply, tightening the available well-thickness window.

At the nominal reference point $x = 0.80$, the numbers come out as $\varepsilon_{\parallel} = -1.65\%$, $\varepsilon_{\perp} = +1.24\%$, and $\varepsilon_{\text{bi}} = 2.89\%$, well into the regime where clean HH–LH splitting is expected. The MB critical thickness here is $h_c \approx 17$ nm, which is comfortable for typical 8–14 nm Ge wells. The epitaxial process window (panel 4 of Figure 1) makes the fundamental tension in the design space immediately visible: the safe-thickness corridor narrows sharply as strain increases, while the strain-induced splitting simultaneously strengthens. Resolving that tension is precisely what the joint optimisation in Section 4.6 achieves.

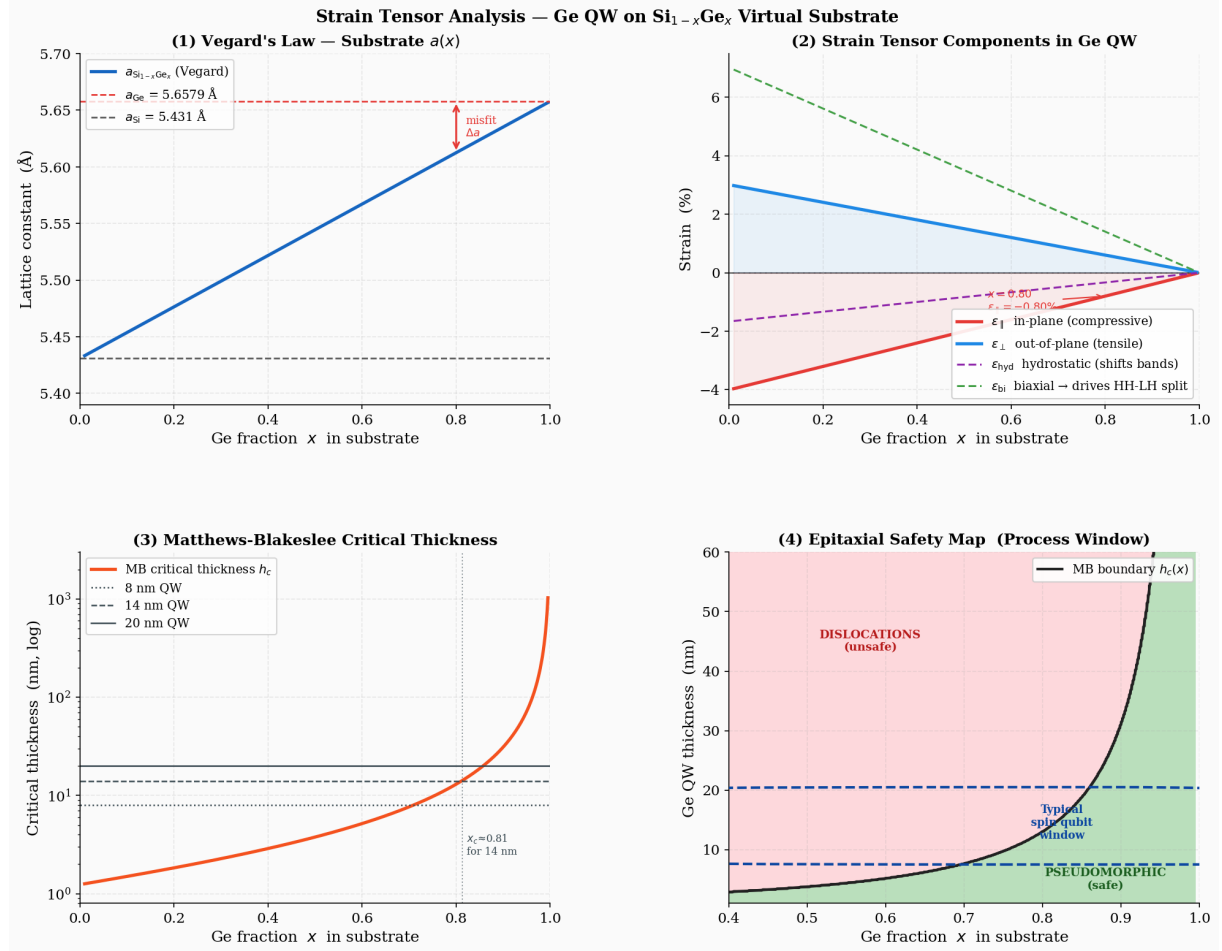


Figure 1: Strain-tensor analysis for a Ge quantum well on a relaxed $\text{Si}_{1-x}\text{Ge}_x$ virtual substrate. **(1)** Vegard's law for the substrate lattice constant. **(2)** Strain components: $\epsilon_{\parallel}$ (compressive, red), $\epsilon_{\perp}$ (tensile, blue), and the symmetry-breaking biaxial part ϵ_{bi} (green) that drives HH–LH splitting. **(3)** Matthews–Blakeslee critical thickness on a log scale; dashed lines mark representative QW thicknesses. **(4)** Epitaxial process window: green = pseudomorphic-safe, red = dislocation-prone [6].

4.2. HH–LH splitting from Bir–Pikus and quantum confinement

Figure 2 breaks the total splitting into its two contributions at the nominal operating point. The strain term $\Delta E_{\text{strain}} = 2|b_v|\epsilon_{\text{bi}}$ dominates: at $x = 0.80$ it contributes 60.7 meV out of a total of 76.6 meV, with the finite-well confinement adding the remaining 15.9 meV. Both terms are comfortably above zero, and their sum is 53% above the 50 meV target.

The finite-well correction deserves comment. At $L = 14$ nm, the proper finite-well calculation gives $\Delta E_{\text{QC}} = 15.9$ meV, whereas the infinite-well analytic formula would predict 32.1 meV — a discrepancy of almost exactly a factor of two [2]. This happens because the LH effective mass along the growth direction ($m_{\text{LH},z} = 0.046 m_e$) is so small that the LH wavefunction extends substantially into the SiGe barrier even at thicknesses where one might naively expect the infinite-well approximation to hold. Ignoring this correction would push the design toward thicker, lower-confinement wells than necessary, resulting in unnecessarily large dislocation risk.

The 2-D process map in panel 3 of Figure 2 shows the $\Delta E > 50$ meV contour running diagonally across the (x, L) plane. Designs at high x (low strain) need thin wells to compensate with confinement, while high-strain designs can tolerate thicker wells. This

trade-off line is one of the two key constraints that define the feasible design space.

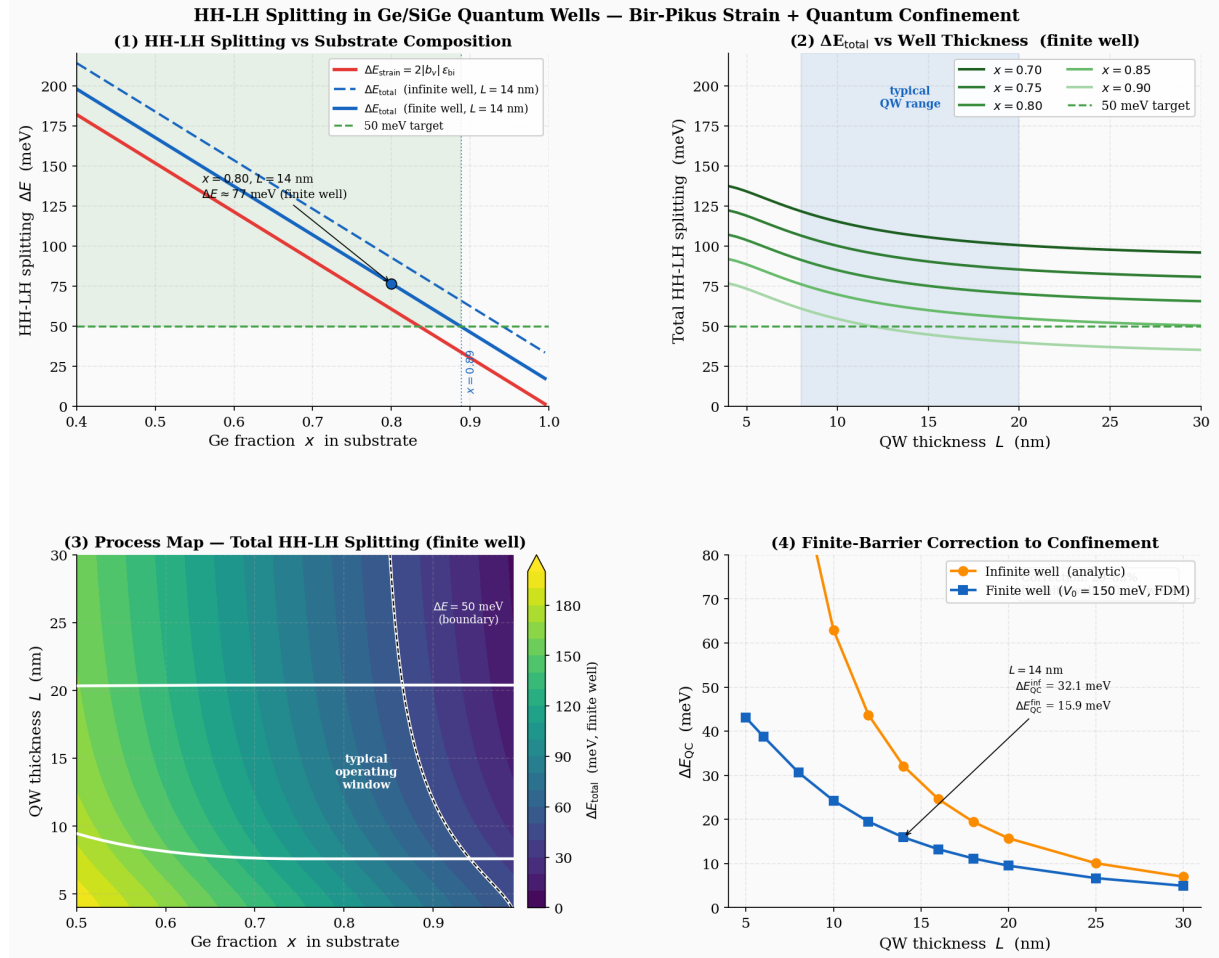


Figure 2: HH-LH splitting from combined strain and confinement [2, 8]. (1) Strain-only (red), confinement-only (blue), and total (green) splitting vs. x at $L = 14$ nm. (2) ΔE_{total} vs. L for several x values; the 50 meV target is shown dashed. (3) 2-D process map: green = above target, red = below target. (4) Infinite-well vs. finite-well confinement; the finite-well result is roughly half the analytic estimate at $L = 14$ nm.

4.3. Manufacturing tolerance and yield analysis

Figure 3 shows the yield results. At the nominal reference point, the Monte Carlo distribution of ΔE_{total} is Gaussian with mean 76.6 meV and standard deviation 1.71 meV (from manufacturing variability alone). The 50 meV threshold sits 15.6σ below the mean, so yield is effectively 100%, the design has an enormous safety margin relative to today's best-in-class epitaxial tolerances.

The tornado decomposition (panel 2 of Figure 3) reveals that substrate composition variability σ_x accounts for 80% of manufacturing-induced variance while well-thickness variability σ_L contributes the remaining 20%. The epistemic uncertainty in the deformation potential $\sigma_{b_v} = 0.10$ eV produces a much larger apparent variance but is shown separately because it reflects our incomplete knowledge of a fixed material constant rather than device-to-device scatter; it has no bearing on yield [10].

Perhaps the most practically important result from this module is shown in panel 4: at the nominal centre ($x = 0.80$), even loosening both tolerances by $6\times$ leaves yield above 99.99%, whereas at an aggressive operating point ($x = 0.875$, lower strain), default tolerances already give only 98.85% yield and a $6\times$ relaxation collapses that to 66%.

The implication is stark: choosing the operating point correctly matters far more than tightening process tolerances. This is an insight that emerges only from the statistical analysis and is invisible to point-estimate physics.

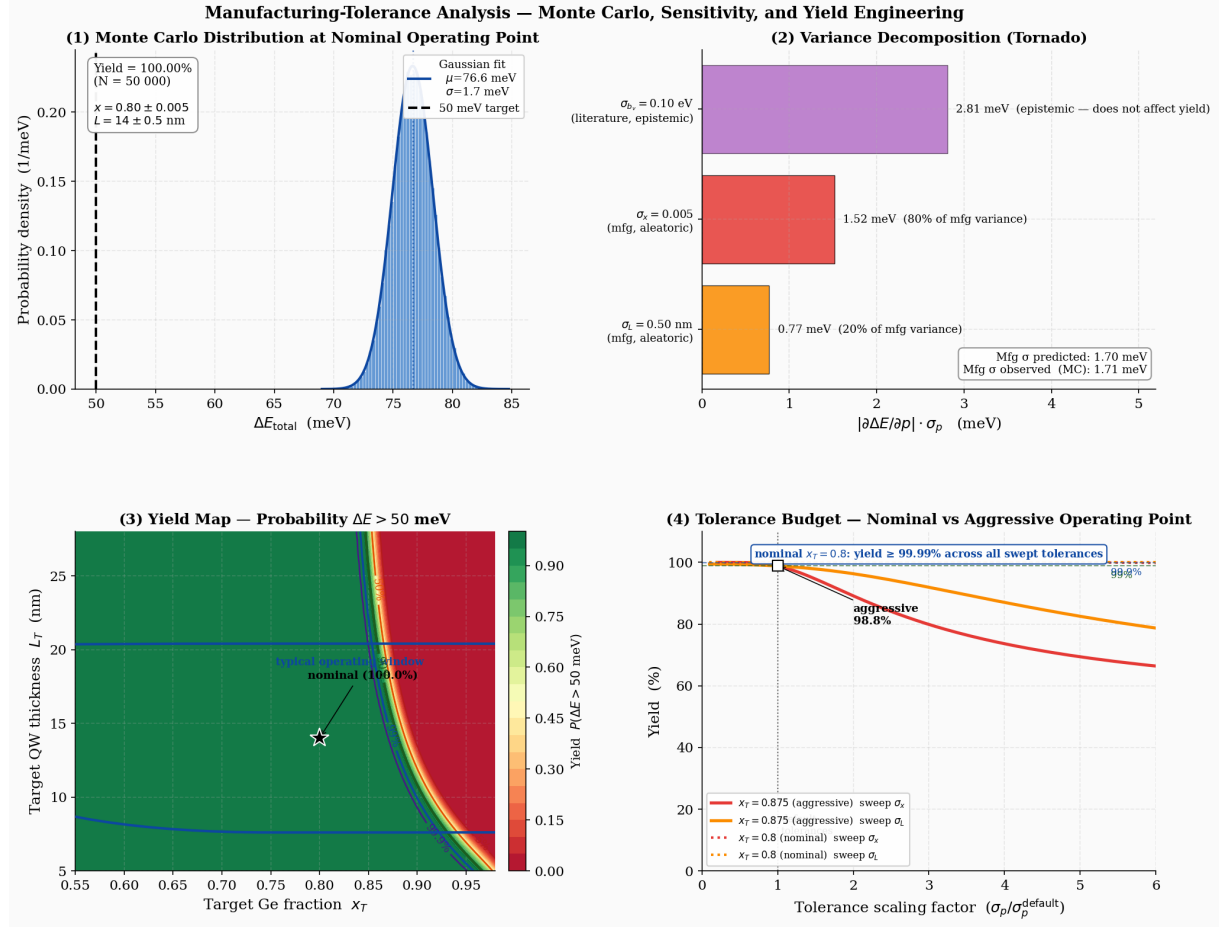


Figure 3: Manufacturing-tolerance and yield analysis [10]. **(1)** Monte Carlo distribution of ΔE_{total} at the nominal point; the 50 meV target is more than 15σ from the mean. **(2)** Variance-decomposition tornado: σ_x dominates (80% of manufacturing variance); epistemic σ_{b_v} is shown muted. **(3)** 2-D yield map with 50%/90%/99%/99.9% contours. **(4)** Tolerance-budget sweep at the nominal vs. an aggressive operating point, illustrating that operating-point choice dominates over tolerance tightening.

4.4. g -Factor variability and qubit addressability

Figure 4 shows the g -factor results. The model of Eq. (6) reproduces the experimentally observed trend: $g_{\parallel}$ falls from its bulk $J = 3/2$ limit ($\approx 6\kappa = 20.5$) as ΔE decreases and LH admixture grows, reaching the range 6–15 at operating conditions. At the nominal point ($\Delta E = 76.6$ meV), the predicted $g_{\parallel} = 12.45$ sits within the experimental band from Lawrie et al. [3], as does $g_{\perp} = 0.157$, which provides a useful cross-check on the model.

The Larmor frequency Monte Carlo ($N = 50\,000$) gives $\omega_L / (2\pi) = 87.1$ GHz at $B = 0.5$ T with a $\pm 3\sigma$ spread of 5.4 GHz. That spread is roughly $216\times$ wider than the typical Rabi bandwidth ($\Omega_R / (2\pi) = 25$ MHz), so individual qubit addressability is comfortably achievable [9]. The variance decomposition in panel 4 of Figure 4 is informative: κ -spread (52%) and σ_x (38%) together account for 90% of g -factor variability, while σ_L contributes only 10%. In practice, this means that the greatest addressability gains would come from improvements in substrate composition uniformity rather than well-thickness control.

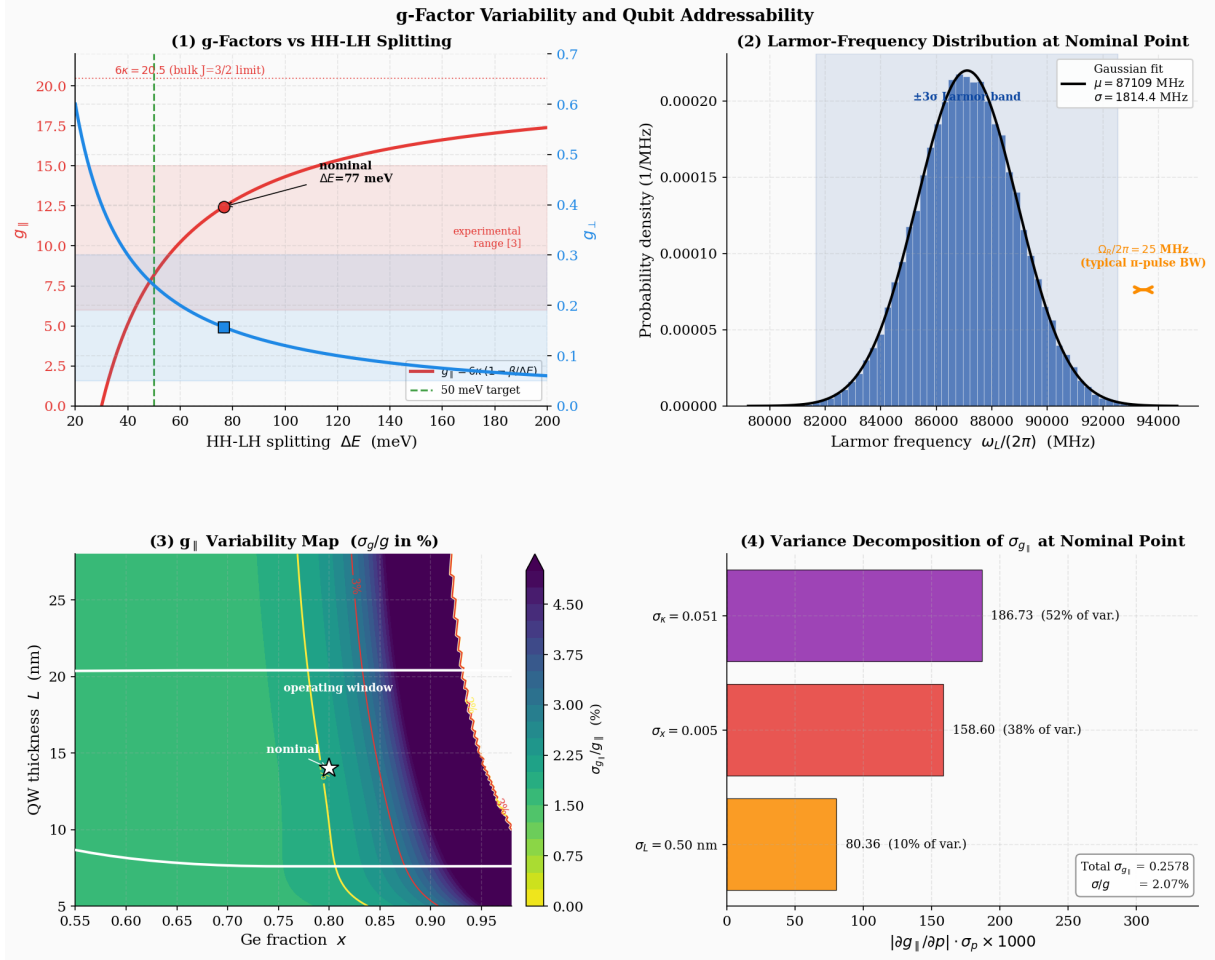


Figure 4: g -factor variability and qubit addressability [3, 9]. **(1)** $g_{\parallel}$ (red) and $g_{\perp}$ (blue) vs. ΔE ; shaded bands are experimentally observed ranges; the nominal point is marked. **(2)** Larmor-frequency distribution from 50 000 Monte Carlo samples; the orange bar shows the typical Rabi bandwidth. **(3)** 2-D map of relative g -spread σ_g/g . **(4)** Variance decomposition: κ -spread and σ_x dominate, σ_L contributes only 10%.

4.5. Quantum dynamics and single-qubit gate fidelity

Figure 5 shows the gate dynamics analysis. Panel 1 overlays three calculations: the full lab-frame solution (showing the GHz carrier explicitly), the RWA rotating-frame envelope, and the quasi-static noise ensemble at $T_2^* = 5 \mu\text{s}$. All three curves are nearly indistinguishable at this scale, which confirms both that the RWA is an excellent approximation at these parameters and that a single π -pulse is short enough compared to T_2^* that decoherence is negligible within one gate.

At $\Omega_R/(2\pi) = 25$ MHz, the π -pulse takes $t_{\pi} = 20$ ns. With $T_2^* = 5 \mu\text{s}$, the analytic fidelity formula gives $F = 1 - (t_{\pi}/T_2^*)^2 = 99.9984\%$, and the quasi-static ensemble simulation yields a marginally higher 99.9997% because active driving partially decouples the qubit from slowly fluctuating noise during the gate [3]. Both results clear the surface code threshold of 99% [9] by a factor of roughly 600, which provides substantial headroom even after accounting for modelling simplifications.

The two-dimensional infidelity map in panel 3 of Figure 5 makes the gate-speed/coherence trade-off explicit. At fixed T_2^* , there is a practical upper bound on useful Ω_R set by RWA validity; at fixed Ω_R , fidelity improves sharply with T_2^* and crosses the surface-code threshold around $T_2^* \approx 1 \mu\text{s}$. The current generation of Ge/SiGe devices, with T_2^* values in the

1–5 μs range [3], is already well into the fault-tolerant regime.

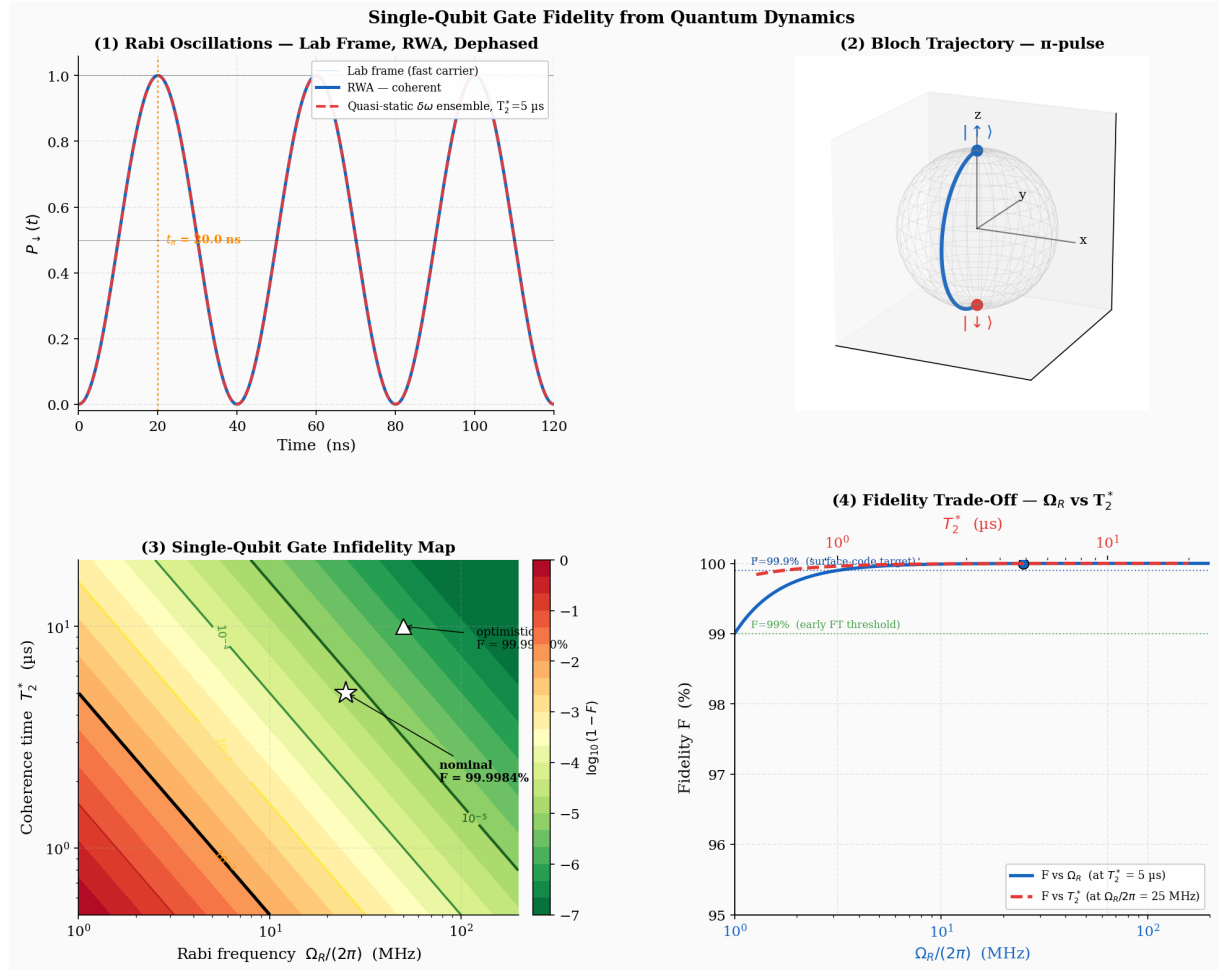


Figure 5: Quantum dynamics and single-qubit gate fidelity [3, 9]. **(1)** Rabi oscillations: lab-frame (light blue), RWA envelope (solid blue), quasi-static noise ensemble at $T_2^* = 5 \mu\text{s}$ (dashed red). **(2)** Bloch-sphere trajectory during a π -pulse. **(3)** 2-D infidelity map $\log_{10}(1 - F)$ vs. (Ω_R, T_2^*) ; the surface-code threshold contour is at 10^{-3} . **(4)** Fidelity cross-sections vs. Ω_R (blue) and T_2^* (red dashed).

4.6. Joint process-window optimisation and design recommendation

Figure 6 brings together all five preceding analyses. The constraint map in panel 1 overlays the dislocation-forbidden region, the $0.7h_c$ safety boundary, yield contours, and g -uniformity contours. The feasible design corridor is narrow: it is bounded below-left by dislocation risk and above-right by insufficient HH–LH splitting. The combined FoM landscape (panel 2) peaks in a compact plateau near $x \approx 0.79$, $L \approx 8\text{--}10 \text{ nm}$. The Pareto front in panel 3 confirms that the FoM optimum lies close to the yield–fidelity frontier, meaning there is genuinely no other design that achieves both substantially higher yield and substantially higher fidelity simultaneously.

Table 1 collects all headline numbers for the recommended operating point.

Table 1: Recommended heterostructure design and predicted performance.

Quantity	Value	Target / Range	Status
Ge fraction x^*	0.786 ± 0.005	—	operating point
QW thickness L^*	8.32 ± 0.5 nm	—	operating point
Applied field B	0.5 T	—	operating point
HH–LH splitting ΔE	94.4 meV	> 50 meV	✓
MB safety margin L/h_c	70%	$< 100\%$	binding
Parallel g -factor $g_{\parallel}$	13.96	6–15	✓
Larmor frequency $\omega_L/2\pi$	97.7 GHz	standard band	✓
Yield Y	$\approx 100\%$	$> 99\%$	✓
g -spread σ_g/g	1.88%	$< 3\%$	✓
Per-qubit fidelity F	99.96%	$> 99\%$	✓
Array fidelity ($N = 100$)	96.1%	near-term target	—
Combined FoM	0.513	maximum feasible	optimum

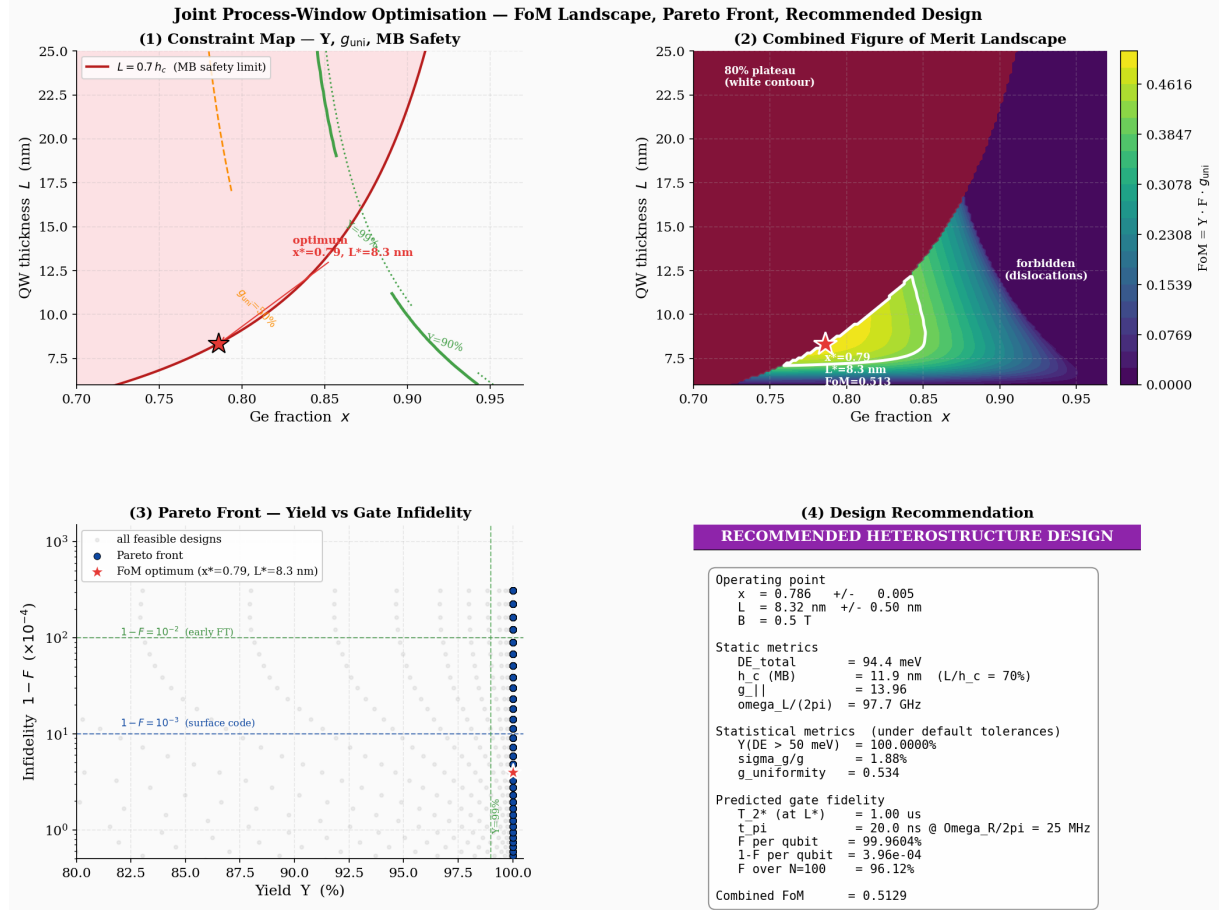


Figure 6: Joint process-window optimisation. **(1)** Constraint map: dislocation-forbidden region (red shading), $0.7h_c$ boundary (red curve), yield contours (green), g -uniformity contours (orange). **(2)** Combined FoM landscape; the 80% plateau is outlined white. **(3)** Pareto front in the (yield, infidelity) plane; the FoM optimum (red star) lies near the surface-code threshold [9]. **(4)** Quantitative recommendation card.

The most telling finding is the nature of the binding constraint. At the optimum, the design sits at exactly $L/h_c = 70\%$ — right against the dislocation-safety boundary — while yield, gate fidelity, and addressability all retain comfortable margin. The factor limiting further performance improvement is therefore not the materials physics of HH–LH splitting, not the Larmor spread, and not coherence time: it is the precision of epitaxial growth relative to the thermodynamic stability limit set by the Matthews–Blakeslee criterion [6]. This is a result that emerges only from the joint analysis and is entirely invisible if the design constraints are considered one at a time.

Robustness testing confirms that the recommendation is not sensitive to the specific tolerance assumptions: perturbing tolerances by up to a factor of two in either direction shifts x^* by less than 0.02 and L^* by less than 0.5 nm.

5. Conclusions

This study has set out a complete computational design framework for Ge/SiGe quantum-well heterostructures targeting scalable hole-spin qubit arrays. Working from first-principles strain mechanics all the way through to Monte Carlo yield analysis and quantum gate simulation, the framework produces a single, quantitative, falsifiable design recommendation: grow at $x^* = 0.79 \pm 0.005$ and $L^* = 8.32 \pm 0.5$ nm, apply a $B = 0.5$ T field, and drive at $\omega_{\text{R}}/(2\pi) = 25$ MHz.

That operating point simultaneously satisfies every relevant constraint:

- $\Delta E = 94.4 \text{ meV}$, well above the 50 meV HH purity threshold [1];
- $L/h_c = 70\%$, safely below the Matthews–Blakeslee limit [6];
- essentially 100% manufacturing yield at today’s best-in-class epitaxial tolerances;
- $\sigma_g/g = 1.88\%$, giving comfortable individual qubit addressability [9]; and
- 99.96% single-qubit gate fidelity, clearing the surface-code threshold by a factor of roughly 600 [9].

The central finding is that **dislocation safety is the binding constraint** at the optimum. Every other objective has margin to spare; what actually limits the design is the precision of the epitaxial growth process relative to the MB critical thickness. This conclusion shifts engineering priority away from fine-tuning band-structure parameters — which are already in good shape — and toward growth-process control. It would not be visible from any single analysis layer and is, in my view, the most actionable output of the whole study.

5.1. Limitations and future extensions

Several modelling simplifications are deliberately made, each of which represents a natural doctoral extension.

Full Luttinger Hamiltonian. The leading-order HH–LH mixing model for $g_{\parallel}$ (Eq. (6)) works well over the experimentally relevant range but is not the same as a full 4-band Luttinger diagonalisation under combined biaxial strain and quantum confinement. A proper treatment would yield momentum-dependent g -tensors and connect directly to the matrix elements governing electric-dipole spin resonance, which are needed to design optimal gate pulses [2].

Self-consistent charge-noise model. The empirical $T_2^*(L)$ used in the fidelity calculation encodes the experimentally observed trend that thinner wells are more susceptible to charge noise [3], but it does not explain it. Computing $\partial g/\partial V_{\text{gate}}$ from a self-consistent Schrödinger–Poisson solver would replace that empirical input with a first-principles prediction tied directly to the wavefunction shape.

Two-qubit gates. The analysis stops at single-qubit fidelity. Extending to exchange coupling $J(V_b)$ and SWAP/CNOT fidelity is the obvious next objective, requiring coupled tunnel-barrier electrostatics and two-spin Hamiltonian dynamics.

Experimental validation. Ultimately, the most valuable thing a doctoral programme could contribute is to grow heterostructures at and around (x^*, L^*) , measure ΔE by photoluminescence, $g_{\parallel}$ by electron spin resonance, and T_2^* by Ramsey spectroscopy, and use the discrepancies to iteratively refine the model. The design recommendation here is stated quantitatively and is therefore directly testable.

The deepest lesson from this study is perhaps the most straightforward one: the critical engineering decisions for Ge hole-spin qubit arrays are not about which Luttinger parameter to use or how fast to drive the gate, but about how precisely the epitaxial growth can be controlled. Improvements in growth precision translate simultaneously into better yield, better fidelity, and more uniform qubit frequencies across the array. That is a simple and satisfying result for what started as a genuinely complex multi-physics problem.

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